

MATHEMATICAL REASONING



SYNOPSIS

In a mathematical language, there are two kinds of reasoning inductive and deductive. We have already discussed the inductive reasoning in the context of mathematical induction. In this chapter, we shall discuss some fundamentals of deductive reasoning.

- **Sentence:** We communicate our ideas or thoughts with the help of sentences in particular languages. Following types of sentences are normally used.

(i) **Assertive Sentence :** A sentence that makes an assertion is called an 'assertive sentence or a declarative sentence'.

Eg: New Delhi is the capital of India

(ii) **Imperative Sentence:** A sentence that expresses a request or a command is called an imperative sentence.

Eg: please give me a glass of water

(iii) **Exclamatory Sentence:** A sentence that expresses some strong feeling is called an exclamatory sentence.

Eg: Oh God! what a beautiful scene

(iv) **Interrogative sentence :** A sentence that asks some question is called an interrogative sentence.

Eg: To which state do you belong?

(v) **Optative sentence:** A sentence that expresses a wish is called an optative sentence.

Eg: God bless you.

- **Statement (or) Proposition :** A sentence is called a mathematically acceptable statement if it is either true(T) or false(F) but not both.

Eg: Natural numbers are always positive
Statements are usually denoted by the letters $p, q, r, \dots$ etc.

- The truthness or falsity of a statement is called its truth value. Truthness of a statement is denoted by T. while its falsity is denoted by F.

- **True statements :** Eg: (i) 2012 is a leap year, (ii) The sum of all interior angles of a triangle is 180° .

- **False statements :**

Eg: (i) All prime numbers are odd integers.
(ii) Two plus two is five.

- **Not a statement :** Eg: (i) Mathematics is difficult. (ii) Tomorrow is Sunday.

- **Simple Statement :** Any statement or proposition whose truth value does not explicitly depend on another statement is called a simple statement.

Eg: Sun rises in the east. Its truth value is T

- **Compound statement:** A statement which is made up of two or more simple statements using the connectives "and($\wedge$)", "or($\vee$)", "implies($\Rightarrow$)", "if and only if($\Leftrightarrow$)" etc... is called a compound statement. In this case each statement is called a component statement.

Eg: This book is for mathematics and its target is Jee-mains

- **Sub-Statement:** The simple statements which form a compound statement are known as its sub-statements or components or constituents.

- If $p, q, r, \dots$ are sub-statements of a compound statement S then the compound statement can be written as $S(p, q, r, \dots)$.

- Compound statement is that its truth value is completely determined by the truth values of the sub-statements together with the way in which they are connected to form the compound statement.

- **Open Statement:** A sentence which contains one or more variables such that when certain values are given to the variable it becomes a statement, is called an open statement.

Eg: "He is a great man" is an open sentence because in the sentence "He" can be replaced by any person.

- **Eg:** Which of the following statement is/are open statement(s)? (1) Ram eats a mango. (2) Krishna goes to school (3) He lives in India (4) Anil and Anuj are good friends.

Sol: In a given options, only option (3) is an open statement, because in this sentence 'he' can be specified to any person.

- **Truth Table :** A table that shows the relationship between the truth value of compound statement, $S(p, q, r, \dots)$ and the truth values of its substatements $p, q, r, \dots$ etc., is called the truth tables of statement S.

- (i) For a single statement p , number of rows $= 2^1 = 2$

P
T
F

- (ii) For two statements p and q, number of rows = $2^2 = 4$

p	q
T	T
T	F
F	T
F	F

- (iii) For the three statements p, q, r,
Number of Rows = $2^3 = 8$

p	q	r
T	F	F
T	F	T
T	T	F
T	T	T
F	F	F
F	F	T
F	T	F
F	T	T

- **Note:** If a compound statement has simply n substatements, then there are 2^n rows representing logical possibilities.

W.E-1: If there are 6 simple statements, then for making a table, find number of rows

Sol: We know that, if compound statements has n substatements, then there are 2^n rows in a table. Here, $n=6$ $\therefore$ Total number of rows = $2^6 = 64$

- **Basic logical connectives or logical operators :** Two or more statements are combined to form a compound statement by using symbols. These symbols are called logical connectives. Logical connectives are given below.

Connective	Symbol	Nature of the compound statement formed by using the connective
and	$\wedge$	Conjunction
or	$\vee$	Disjunction
If then	$\Rightarrow$ or $\rightarrow$	Implication or conditional
If and only if (iff)	$\Leftrightarrow$ or $\leftrightarrow$	Equivalence or Bi-conditional or Bi-implication
not	$\sim$	Negation

- **Negation ($\sim$) :** The process of forming the contradictory of a given statement is called negation.
- If p is a statement, then the negation of p is also a statement denoted by $\sim p$.
- Eg :- p: New Delhi is a city, then $\sim p$: It is false that New Delhi is a city. (or) New Delhi is not a city

- **Negation Truth table :**

p	$\sim p$
T	F
F	T

- **Conjunction ($\wedge$) :** Any two simple statements can be connected by the word 'and' to form a compound statement called the conjunction of the original statements.

Let p and q be two statements. The conjunction of p and q is denoted by $p \wedge q$, read as p and q

- **Truth table for Conjunction :**

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

- **Eg:-** p: Two is an even number.
q: Two is a prime number.
 $p \wedge q$: Two is an even number and a prime number.

- **Disjunction (Alternation) ($\vee$) :** Any two statements can be connected by the word 'or' to form a compound statement called the 'disjunction' of the original statements.

Let p and q be two statements the disjunction of p and q is denoted by $p \vee q$, read as p or q.

- **Truth table for disjunction :**

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

- Eg: **p**: Two is an even number.
q: Two is a prime number.
 $p \vee q$: Two is an even number or a prime number.
- **Conditional (or) Implication**: Two statements connected by the connective phrase 'if ... then' give rise to a compound statement which is known as an implication or a conditional statement.
If p and q are two statements forming the implication of 'if p then q ' then the implication denoted by ' $p \Rightarrow q$ ' or ' $p \rightarrow q$ '. p is called the 'antecedent' and q is called 'consequent'.
➤ $p \Rightarrow q$ read as p implies q , q if p , p is sufficient for q , q is necessary for p .
➤ $p \Rightarrow q$ is the statement that is false when p is true and q is false and true otherwise.
- **Truth table for Conditional :**

p	q	$p \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

- Eg:- **p**: An integer is a multiple of 9.
q: An integer is a multiple of 3.
 $p \rightarrow q$: If an integer is a multiple of 9 then it is a multiple of 3.
- **Bi-implication** ($\Leftrightarrow$ (or) $\Leftrightarrow$): A statement is a biconditional statement if it is the conjunction of two conditional statements one converse to the other. If p , q are two statements, then the compound statement $p \Rightarrow q$ and $q \Rightarrow p$ is called a biconditional statement or an equivalence and is denoted by $p \Leftrightarrow q$, $(p \Rightarrow q) \wedge (q \Rightarrow p)$.
➤ $p \Leftrightarrow q$ is the statement that is true when p and q have the same truth value and otherwise false.
- **Truth table for Bi-implication :**

p	q	$p \Leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

- Eg:- **p**: A number is divisible by 3.
q: Sum of the digits of a number is divisible by 3.
 $p \Leftrightarrow q$: A number is divisible by 3 if and only if the sum of its digits is divisible by 3.
- **Converse**: Let p, q be two statements. "If q then p " is called the converse of "if p then q ".
Thus the converse of $p \Rightarrow q$ is $q \Rightarrow p$.
- **Inverse**: Let p, q be two statements. "if $\sim p$ then $\sim q$ " is called the inverse of "if p then q ". Thus the inverse of $p \Rightarrow q$ is $\sim p \Rightarrow \sim q$.
- **Contrapositive**: The statement $\sim q \rightarrow \sim p$ is called the contrapositive of $p \rightarrow q$.
- Eg:- **p**: x is an even integer. **q**: x^2 is divisible by 4.
(i) $p \rightarrow q$: If x is even integer then x^2 is divisible by 4.
(ii) $q \rightarrow p$: x^2 is divisible by 4 then x is even.
(iii) $\sim p \rightarrow \sim q$: If x is not even integer then x^2 is not divisible by 4.
(iv) $\sim q \rightarrow \sim p$: if x^2 is not divisible by 4 then x is not an even integer.
- **Converse, inverse and contra positive of a conditional**: Suppose p, q are two statements such that $p \Rightarrow q$ then
i) Converse is $q \Rightarrow p$
ii) Inverse is $\sim p \Rightarrow \sim q$
iii) Contra positive is $\sim q \Rightarrow \sim p$
- **Truth Table :**

p	q	Conditional $p \Rightarrow q$	Converse $q \Rightarrow p$	Inverse $\sim p \Rightarrow \sim q$	Contra positive $\sim q \Rightarrow \sim p$
T	T	T	T	T	T
T	F	F	T	T	F
F	T	T	F	F	T
F	F	T	T	T	T

- **Tautology, Contradiction**: (i) A compound statement that is **always true** is called a tautology.
(ii) A compound statement that is **always false** is called a contradiction or fallacy.
Eg:- Let p be a statement
- **Truth table :**

P	$\sim P$	$p \vee \sim P$	$p \wedge \sim P$
T	F	T	F
F	T	T	F

- (i) $p \vee \sim p$ is a tautology
(ii) $p \wedge \sim p$ is a contradiction

➤ **Logical Equivalence:** The statements **p** and **q** are called logically equivalent if they have the same entries in the last column of the truth tables.

➤ **Eg:-** (i) $(p \vee q)$ and $(\sim p) \wedge (\sim q)$ are logically equivalent.

(ii) $p \rightarrow q$ and $\sim p \vee q$ are logically equivalent.

(iii) $p \Rightarrow q \equiv \sim q \Rightarrow \sim p$

(iv) $\sim(p \rightarrow q) \equiv p \wedge \sim q$

p	q	$p \vee q$	$\sim(p \vee q)$	$\sim p$	$\sim q$	$\sim p \wedge \sim q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

p	q	$\sim p$	$(\sim p) \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

- The phrases, 'for all', 'for some', 'for no', 'for every' and 'there exists atleast one' convey the idea of quantity and refer to some specific collection of numbers or objects. these phrases quantify the variable in open sentences. they are called quantifiers.
- The quantifier 'for all' or 'for every' is called the universal quantifier and is denoted by $\forall$. The quantifier 'some' or 'there exists atleast one' is called existential quantifier and is denoted by the symbol $\exists$.

➤ **Algebra of statements :**

- Idempotent laws :** For any statement **p**,
(i) $p \vee p \equiv p$ (ii) $p \wedge p \equiv p$
- Commutative laws:** For any statements **p** and **q**
(i) $p \vee q \equiv q \vee p$ (ii) $p \wedge q \equiv q \wedge p$
- Associative laws:** For any three statements **p, q, r**
(i) $(p \vee q) \vee r \equiv p \vee (q \vee r)$ (ii) $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
- Distributive Laws:** For any three statements **p, q, r**,
(i) $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
(ii) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

5. **DeMorgan's laws:** If **p** and **q** are two statements, then

(i) $\sim(p \wedge q) \equiv (\sim p) \vee (\sim q)$

(ii) $\sim(p \vee q) \equiv (\sim p) \wedge (\sim q)$

6. **Identity laws:** If **t** and **c** denote a tautology and a contradiction respectively, then for any statement **p**,

(i) $p \wedge t \equiv p$

(ii) $p \vee c \equiv p$

(iii) $p \vee t \equiv t$

(iv) $p \wedge c \equiv c$

7. **Complement laws:** For any statement **p**,

(i) $p \vee (\sim p) \equiv t$

(ii) $p \wedge (\sim p) \equiv c$

8. **Law of contrapositive :** For any two statements **p** and **q**,

(i) $p \Rightarrow q \equiv \sim q \Rightarrow \sim p$

9. **Involution Laws:** For any statement **p**, we have $\sim(\sim p) \equiv p$.



1. If a sentence can be judged to be true or false, but not both then it is called

- 1) an open sentence
- 2) a statement
- 3) a tautology
- 4) a contradiction

2. The truthfulness or falsity of a statement is called its

- 1) negation
- 2) converse
- 3) inverse
- 4) truth value

3. Which of the following is a proposition

- 1) Logic is an interesting subject
- 2) He is very talented
- 3) I am a lion
- 4) A triangle is a circle and 10 is a prime number

4. Which of the following is not a proposition

- 1) 3 is a prime
- 2) Mathematics is interesting
- 3) 5 is an even integer
- 4) $\sqrt{2}$ is irrational

5. Denial of a statement is called is

- 1) negation
- 2) converse
- 3) inverse
- 4) truth value

6. The symbolic form of "p or q" is

- 1) $p \vee q$
- 2) $p \wedge q$
- 3) $p \Rightarrow q$
- 4) $p \Leftrightarrow q$

7. The symbolic form of "p and q" is

- 1) $p \vee q$
- 2) $p \wedge q$
- 3) $p \Rightarrow q$
- 4) $p \Leftrightarrow q$

8. The symbolic form of "(either p) or (not p)" is

- 1) $p \vee (\sim p)$
- 2) $p \wedge (\sim p)$
- 3) $p \Rightarrow (\sim p)$
- 4) $p \Leftrightarrow (\sim p)$

9. The symbolic form of “p only if q” is
 1) $p \vee q$ 2) $p \wedge q$ 3) $p \Rightarrow q$ 4) $q \Rightarrow p$
10. $p : x$ is odd ; $q : x^2$ is odd. The symbolic form of “x is odd and x^2 is not odd”, is
 1) $p \vee q$ 2) $p \wedge q$ 3) $(\sim p) \vee q$ 4) $p \wedge (\sim q)$
11. $p : x$ is odd ; $q : x^2$ is odd. the symbolic form of “If x is not odd then x^2 is odd”, is
 1) $p \Rightarrow q$ 2) $q \Rightarrow p$
 3) $(\sim p) \Rightarrow q$ 4) $(\sim q) \Rightarrow p$
12. The converse of $p \Rightarrow q$ is
 1) $p \Rightarrow q$ 2) $q \Rightarrow p$
 3) $\sim p \Rightarrow \sim q$ 4) $\sim q \Rightarrow \sim p$
13. The inverse of $p \Rightarrow q$ is
 1) $p \Rightarrow q$ 2) $q \Rightarrow p$
 3) $\sim p \Rightarrow \sim q$ 4) $\sim q \Rightarrow \sim p$
14. The contrapositive of $p \Rightarrow q$ is
 1) $p \Rightarrow q$ 2) $q \Rightarrow p$
 3) $\sim p \Rightarrow \sim q$ 4) $\sim q \Rightarrow \sim p$
15. $p \wedge (\sim p) \Rightarrow p$ is
 1) a tautology 2) a contradiction
 3) a tautology and a contradiction
 4) neither a tautology nor a contradiction
16. $p \wedge (\sim p)$ is
 1) a tautology 2) a contradiction
 3) a tautology and a contradiction
 4) neither a tautology nor a contradiction
17. $p \vee (\sim p)$ is
 1) a tautology 2) a contradiction
 3) a tautology and a contradiction
 4) neither a tautology nor a contradiction
18. $p \Rightarrow p \vee q$ is
 1) a tautology 2) a contradiction
 3) a tautology and a contradiction
 4) neither a tautology nor a contradiction
19. Which of the following is the universal quantifier?
 1) $\in$ 2) $\subset$ 3) $\forall$ 4) $\exists$
20. The quantifier implied in the statement “No student is lazy” is
 1) $\exists$ 2) $\sim$ 3) $\forall$ 4) $\Rightarrow$

CUQ KEY

- | | | | |
|-------|-------|-------|-------|
| 1) 2 | 2) 4 | 3) 4 | 4) 2 |
| 5) 1 | 6) 1 | 7) 2 | 8) 1 |
| 9) 3 | 10) 4 | 11) 3 | 12) 2 |
| 13) 3 | 14) 4 | 15) 1 | 16) 2 |
| 17) 1 | 18) 1 | 19) 3 | 20) 3 |

CUQ HINTS

15) $p \wedge (\sim p)$ is always contradiction.

$\therefore p \wedge (\sim p) \Rightarrow p$ is a tautology.

17) $p \vee (\sim p)$ is always a tautology.


LEVEL-I (C.W)
CONJUNCTION - DISJUNCTION

1. The disjunction of the statement, “ It is raining”; The sun is shining”, is
 1) It is raining and the sun is shining
 2) It is raining or the sun is shining
 3) It is raining and the sun is not shining
 4) It is not raining or the sun is not shining
2. “5 + 7 = 10 and 4 + 3 = 7”. Write the statement using the appropriate connective
 1) $5 + 7 = 10 \vee 4 + 3 = 7$
 2) $5 + 7 = 10 \wedge 4 + 3 = 7$
 3) $5 + 7 = 10 \Rightarrow 4 + 3 = 7$
 4) $5 + 7 = 10 \Leftrightarrow 4 + 3 = 7$
3. “7 is odd or 7 is prime”. Write the statement using the appropriate connective
 1) 7 is odd $\vee$ 7 is prime 2) 7 is odd $\wedge$ is prime
 3) 7 is odd $\Rightarrow$ 7 is prime
 4) 7 is odd $\Leftrightarrow$ 7 is prime

IMPLICATION & BI-IMPLICATION

4. The truth value of “if 3 + 2 = 5 then 1 x 0 = 0” is
 1) T 2) F 3) T or F 4) T and F
5. p: he is hard working q: he will win. The symbolic form of “if he is hard working then he will win”, is
 1) $p \vee q$ 2) $p \wedge q$
 3) $p \Rightarrow q$ 4) $q \Rightarrow p$

**NEGATION, INVERSE, CONVERSE,
CONTRAPOSITIVE, TAUTOLOGY,
CONTRADICTION**

6. "Earth is a planet", the negation of this statement is

- 1) The earth is round
- 2) The earth is not round
- 3) The earth revolves round the sun
- 4) The earth is not a planet

7. The truth table of $(\sim p) \wedge q$ is

1)

p	q	$(\sim p) \wedge q$
T	T	F
T	F	F
F	T	T
F	F	F

2)

p	q	$(\sim p) \wedge q$
T	T	F
T	F	T
F	T	F
F	F	F

3)

p	q	$(\sim p) \wedge q$
T	T	F
T	F	F
F	T	F
F	F	F

4)

p	q	$(\sim p) \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

8. The truth table of $(\sim p) \Rightarrow q$ is

1)

p	q	$(\sim p) \Rightarrow q$
T	T	F
T	F	T
F	T	T
F	F	T

2)

p	q	$(\sim p) \Rightarrow q$
T	T	T
T	F	T
F	T	T
F	F	F

3)

p	q	$(\sim p) \Rightarrow q$
T	T	F
T	F	T
F	T	F
F	F	T

4)

p	q	$(\sim p) \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

9. Negation of the compound proposition: If the examination is difficult, then I shall pass if I study hard

- 1) The examination is difficult and I study hard and I shall pass
- 2) The examination is difficult and I study hard but I shall not pass
- 3) The examination is not difficult and I study hard and I shall pass
- 4) All of these

10. Which of the following is not logically equivalent to the proposition : “A real number is either rational or irrational”

- 1) If a number is neither rational nor irrational then it is not real
- 2) If a number is not a rational or not an irrational, then it is not real
- 3) If a number is not real, then it is neither rational nor irrational
- 4) If a number is real, then it is rational or irrational

11. If $p \rightarrow (q \vee r)$ is false, then the truth values of p, q, r are respectively

- 1) T, F, F 2) F, F, F 3) F, T, T 4) T, T, F

12. The proposition $(p \rightarrow \sim p) \wedge (\sim p \rightarrow p)$ is

- 1) a tautology
- 2) a contradiction
- 3) neither a tautology nor a contradiction
- 4) a tautology and a contradiction

13. Let p be the proposition: Mathematics is interesting and let q be the proposition that Mathematics is difficult, then the symbol $p \wedge q$ means

- 1) Mathematics is interesting implies that Mathematics is difficult
- 2) Mathematics is interesting implies and is implied by Mathematics is difficult
- 3) Mathematics is interesting and Mathematics is difficult
- 4) Mathematics is interesting or Mathematics is difficult

14. The negation of the compound proposition $p \leftrightarrow \sim q$ is logically equivalent to

- 1) $p \leftrightarrow q$
- 2) $(p \rightarrow q) \wedge (\sim q \rightarrow p)$
- 3) $(\sim q \rightarrow p) \vee (\sim p \rightarrow q)$
- 4) $(\sim p \wedge \sim q) \vee (q \wedge \sim p)$

LEVEL-I (C.W.) - KEY

- 1) 2 2) 2 3) 1 4) 1 5) 3
- 6) 4 7) 1 8) 2 9) 2 10) 2
- 11) 1 12) 2 13) 3 14) 1

LEVEL-I (C.W.) - HINTS

11) Verify the options

14) Apply the formula $\sim (p \leftrightarrow q) \equiv p \leftrightarrow \sim q$



LEVEL-I (H.W)

1. The disjunction of the statement, “Hyderabad is in India”, “ $2+2=4$ ”, is

- 1) Hyderabad is in India and $2+2=4$
- 2) Hyderabad is in India or $2+2=4$
- 3) Hyderabad is in India but not $2+2=4$
- 4) Hyderabad is not in India or $2+2=4$

2. The disjunction of the statements, “ $1+2=3$ ”; “ $3+4=6$ ”, is

- 1) $1+2=3$ and $3+4=6$
- 2) $1+2=3$ or $3+4=6$
- 3) $1+2=3$ but not $3+4=6$
- 4) neither $1+2=2$ or $3+4=6$

3. The truth value of “ $20+10=2$ and $20 \times 10 = 200$ ” is

- 1) T 2) F 3) T or F 4) T and F

4. The Truth value of “if $5 \times 7 = 30$ then $2+1=4$ ” is

- 1) T 2) F 3) T or F 4) T and F

5. Which of the following is true?

- 1) $3+7=10 \Leftrightarrow 1+2=3$ 2) $3+7=10 \Leftrightarrow 1+2=2$
- 3) $3 \times 7=10 \Leftrightarrow 1 \times 3=3$ 4) $3 \times 7=8 \Leftrightarrow 1 \times 2=2$

6. The truth table of $(\sim p) \wedge (\sim q)$

1)

p	q	$(\sim p) \wedge (\sim q)$
T	T	F
T	F	F
F	T	T
F	F	F

2)

p	q	$(\sim p) \wedge (\sim q)$
T	T	F
T	F	T
F	T	F
F	F	F

3)

p	q	$(\sim p) \wedge (\sim q)$
T	T	F
T	F	F
F	T	F
F	F	T

4)

p	q	$(\sim p) \wedge (\sim q)$
T	T	T
T	F	F
F	T	F
F	F	F

7. The truth table of $p \Rightarrow \sim q$ is

1)

p	q	$p \Rightarrow \sim q$
T	T	F
T	F	T
F	T	T
F	F	T

2)

p	q	$p \Rightarrow \sim q$
T	T	T
T	F	T
F	T	T
F	F	F

3)

p	q	$p \Rightarrow \sim q$
T	T	F
T	F	T
F	T	F
F	F	T

4)

p	q	$p \Rightarrow \sim q$
T	T	T
T	F	F
F	T	T
F	F	T

8. Negation of the proposition : If we control population growth, we prosper.

- 1) If we do not control population growth, we prosper
- 2) If we control population, we do not prosper
- 3) We control population but we do not prosper
- 4) We do not control population, but we prosper

9. The negation of the proposition "If 2 is prime, then 3 is odd" is

- 1) If 2 is not prime then 3 is not odd
- 2) 2 is prime and 3 is not odd
- 3) 2 is not prime and 3 is odd
- 4) If 2 is not prime then 3 is odd

LEVEL-I (H.W.) - KEY

- 1) 2 2) 2 3) 2 4) 1 5) 1
6) 3 7) 1 8) 3 9) 2



LEVEL-II - (C.W)

CONJUNCTION & DISJUNCTION

1. Let there be two propositions
p: I take only bread and butter in breakfast.
q: I do not take any thing in breakfast.
Then the compound proposition "I take only bread and butter in breakfast or I do not take any thing" is represented by
1) $p \wedge q$ 2) $p \vee q$ 3) $p \rightarrow q$ 4) $p \leftrightarrow q$

NEGATION, INVERSE, CONVERSE, CONTRAPOSITIVE, TAUTOLOGY, CONTRADICTION

2. The negation of the proposition: "If a number is divisible by 15, then it is divisible by 5 or 3:."
 - 1) If a number is divisible by 15, then it is not divisible by 5 and 3.
 - 2) A number is divisible by 15, and it is not divisible by 5 and 3.
 - 3) A number is divisible by 15, then it is not divisible by 5 or 3.
 - 4) A number is not divisible by 15 or it is not divisible by 5 and 3.
3. Negation of the statement $\sim p \rightarrow (q \vee r)$
 - 1) $\sim p \rightarrow (q \vee r)$ 2) $p \vee (q \wedge r)$
 - 3) $\sim p \wedge (\sim q \wedge \sim r)$ 4) $p \wedge (q \vee r)$
4. The converse of "if in a triangle ABC, $AB = AC$ then $\angle B = \angle C$ " is
 - 1) If in a triangle ABC, $\angle B = \angle C$ then $AB = AC$
 - 2) If in a triangle ABC, $AB \neq AC$, then $\angle B \neq \angle C$
 - 3) If in a triangle ABC, $\angle B \neq \angle C$ then $AB \neq AC$
 - 4) If in a triangle ABC, $\angle B \neq \angle C$, then $AB = AC$
5. The inverse of the proposition $(p \wedge \sim q) \rightarrow r$ is
 - 1) $\sim r \rightarrow p \vee q$ 2) $\sim p \vee q \rightarrow \sim r$
 - 3) $r \rightarrow p \wedge \sim q$ 4) $\sim p \rightarrow (p \wedge r)$

6. The contrapositive of “if two triangles are congruent then they are similar” is
- 1) If two triangles are similar then they are congruent
 - 2) If two triangles are not congruent then they are not similar
 - 3) If two triangles are not similar then they are not congruent
 - 4) If two triangles are similar then they are not congruent
7. “The diagonals of a rhombus are perpendicular”. The contrapositive of the above statement is
- 1) If the figure is not a rhombus, then its diagonals are not perpendicular.
 - 2) If the diagonals are perpendicular, then the figure is a rhombus.
 - 3) If the diagonals are not perpendicular, then the figure is a rhombus
 - 4) If the diagonals are not perpendicular, then the figure is not a rhombus
8. The contrapositive of $(\sim p \wedge q) \rightarrow (q \wedge \sim r)$ is
- 1) $(p \vee \sim q) \rightarrow (\sim q \vee p)$
 - 2) $(\sim q \vee r) \rightarrow (\sim p \vee q)$
 - 3) $(\sim q \vee r) \rightarrow (p \vee \sim q)$
 - 4) $(\sim p \vee r) \rightarrow (\sim p \wedge \sim r)$
9. Consider the following propositions
p: I have the raincoat
q: I can walk in the rain.
 The proposition “If I have the raincoat, then I can walk in the rain” is represented by
- 1) $p \rightarrow q$
 - 2) $q \rightarrow p$
 - 3) $p \rightarrow \sim q$
 - 4) $p \leftrightarrow q$
10. Given that water freezes below zero degree celsius.
 Consider the following statements :
p: Water freeze this morning,
q: This morning temperature was below $0^{\circ}C$.
 Which of the following is correct?

- 1) p and q are logically equivalent
 - 2) p is the inverse of q
 - 3) p is the converse of q
 - 4) p is the contrapositive of q
11. Consider the following proposition
p: I take medicine, q: I can sleep. Then, the compound statement $\sim p \rightarrow \sim q$ means
- 1) If I do not take medicine, then I cannot sleep
 - 2) If do not take medicine, then I can sleep
 - 3) I take medicine iff I can sleep
 - 4) If take medicine if I can sleep.
12. The statement $\sim p \vee q$ is equivalent to
- 1) $p \rightarrow q$
 - 2) $\sim p \rightarrow q$
 - 3) $\sim p \rightarrow \sim q$
 - 4) $p \rightarrow \sim q$
13. If p: 4 is an even prime number
 q: 6 is a divisor of 12 and
 r: the HCF of 4 and 6 is 2, then which one of the following is true ?
- 1) $p \wedge q$
 - 2) $(p \vee q) \wedge \sim r$
 - 3) $\sim (q \wedge r) \vee p$
 - 4) $\sim p \vee (q \wedge r)$

LEVEL-II (C.W.) - KEY

- 1) 2
- 2) 2
- 3) 3
- 4) 1
- 5) 2
- 6) 3
- 7) 4
- 8) 3
- 9) 1
- 10) 1
- 11) 1
- 12) 1
- 13) 4

LEVEL-II (C.W.) - HINTS

- 3) Apply the formula $\sim (p \rightarrow q) \equiv p \wedge \sim q$
 8) Contra positive of $p \Rightarrow q$ is $\sim q \Rightarrow \sim p$



LEVEL-II - (H.W)

1. The negation of the proposition “If 2 is prime then 3 is odd” is
- 1) If 2 is not prime then 3 is not odd
 - 2) 2 is prime and 3 is not odd
 - 3) 2 is not prime and 3 is odd
 - 4) If 2 is not prime then 3 is odd
2. Negation of the statement $p \rightarrow (q \vee r)$ is
- 1) $\sim p \rightarrow \sim (q \vee r)$
 - 2) $\sim p \rightarrow \sim (q \wedge r)$
 - 3) $(q \vee r) \rightarrow p$
 - 4) $p \wedge (\sim q \wedge \sim r)$

3. The inverse of “if x has courage then x will win”, is
 - 1) if x will win then x has courage
 - 2) if x has no courage then x will not win
 - 3) if x will not win then x has no courage
 - 4) if x will not win then x has courage
4. The contrapositive of the statement “if $2^2 = 5$, then I get first class” is
 - 1) If I do not get a first class, then $2^2 = 5$
 - 2) If I do not get a first class, then $2^2 \neq 5$
 - 3) If I get a first class, then $2^2 = 5$
 - 4) If I get a first class, then $2^2 \neq 5$
5. The contrapositive of “if in a triangle ABC , $AB = AC$ then $\angle B = \angle C$ ” is
 - 1) If in a triangle ABC , $\angle B = \angle C$ then $AB = AC$
 - 2) If in a triangle ABC , $AB \neq AC$, then $\angle B \neq \angle C$
 - 3) If in a triangle ABC , $\angle B \neq \angle C$ then $AB \neq AC$
 - 4) If in a triangle ABC , $\angle B \neq \angle C$, then $AB = AC$
6. The contrapositive of $2x + 3y = 9 \Rightarrow x \neq 4$ is
 - 1) $x = 4 \Rightarrow 2x + 3y \neq 9$
 - 2) $x = 4 \Rightarrow 2x + 3y = 9$
 - 3) $x \neq 4 \Rightarrow 2x + 3y \neq 9$
 - 4) $x \neq 4 \Rightarrow 2x + 3y = 9$

LEVEL-II (H.W.) - KEY

- | | | | |
|------|------|------|------|
| 1) 2 | 2) 4 | 3) 2 | 4) 2 |
| 5) 3 | 6) 1 | | |



LEVEL-III

1. Write the negation of the statement if $5 > 7$ then, $6 < 4$
 - 1) $5 > 7 \cap (6 \geq 4)$
 - 2) $5 > 7 \cup (6 \leq 4)$
 - 3) $5 \leq 7 \cup (6 \leq 4)$
 - 4) $5 \leq 7 \cap (6 \leq 4)$
2. The statement $\sim(p \leftrightarrow \sim q)$ is (Mains 2014)
 - 1) a tautology
 - 2) a fallacy
 - 3) equivalent to $p \leftrightarrow q$
 - 4) equivalent to $\sim p \leftrightarrow q$
3. Which one of the following is not a contradiction
 - 1) $[\sim p \wedge (p \vee \sim q)] \wedge q$
 - 2) $\sim(p \wedge q) \wedge p$
 - 3) $[(p \rightarrow q) \rightarrow p] \wedge \sim p$
 - 4) $(\sim p \rightarrow \sim q) \leftrightarrow (p \rightarrow q)$

4. Write down the negation of the statement it is raining and is cool.
 - 1) it is not raining and it is cool
 - 2) it is raining and it is cool
 - 3) neither it is raining nor it is cool.
 - 4) it is raining and it is not cool
5. $[p \wedge (\sim q)] \wedge [(\sim p) \vee q]$ is
 - 1) a tautology
 - 2) a contradiction
 - 3) a tautology and a contradiction
 - 4) neither a tautology nor a contradiction
6. The statement $p \wedge (\sim q \vee r) \vee (\sim r \vee q)$ is
 - 1) tautology
 - 2) contradiction
 - 3) contrapositive
 - 4) neither a tautology nor a contradiction
7. Let S be a non-empty subset of R . Consider the following statement :
P: There is a rational number $x \in S$ such that $x > 0$. Which of the following statements is the negation of the statement P ?
 - 1) There is no rational number $x \in S$ such that $x \leq 0$
 - 2) Every rational number $x \in S$ satisfies $x \leq 0$
 - 3) $x \in S$ and $x \leq 0 \Rightarrow$ is not rational
 - 4) There is a rational number $x \in S$ such that $x \leq 0$
8. $(p \Rightarrow q) \rightarrow [(r \vee p) \Rightarrow (r \vee q)]$ is
 - 1) a contradiction
 - 2) a tautology
 - 3) a tautology and a contradiction
 - 4) neither a tautology nor a contradiction
9. Find the truth set of the open statement
 $p(x) \equiv x^2 - x(3 + \sqrt{2}) + 3\sqrt{2} = 0 \quad \forall x \in Q$
 - 1) $\{\sqrt{2}\}$
 - 2) $\{3\}$
 - 3) $\{3, \sqrt{2}\}$
 - 4) $\{\}$
10. Suppose p : A natural number n is odd and q : natural number n is not divisible by 2, then the biconditional statement $p \Leftrightarrow q$ is
 - 1) A natural number n is odd, if and only if it is divisible by 2
 - 2) A natural number n is odd, if and only if it is not divisible by 2.
 - 3) If a natural number n is odd, then it is not divisible by 2
 - 4) A natural number n is odd, then it is divisible by 2.

11. Write negation of the statement if $\triangle ABC$ is isosceles then the base angles A and B are equal.

- 1) $\triangle ABC$ is isosceles and the base angle A and B are not equal
- 2) $\triangle ABC$ is isosceles and the base angle A and B are equal
- 3) $\triangle ABC$ is not isosceles and the base angle A and B are not equal
- 4) all the above

12. The statement $p \rightarrow (q \rightarrow p)$ is equivalent to

- 1) $p \rightarrow (p \rightarrow q)$
- 2) $p \rightarrow (q \vee p)$
- 3) $p \rightarrow (q \wedge p)$
- 4) $p \rightarrow (p \leftrightarrow q)$

13. Write the converse of the statement if

$x + 2 = 6$, then $x = 4$

- 1) If $x \neq 4$, then $x + 2 = 6$
- 2) If $x = 4$ and $x + 2 = 6$
- 3) If $x + 2 = 6$ then $x = 4$
- 4) If $x = 4$, then $x + 2 = 6$

14. If p, q, r have truth values T, F, T respectively, which of the following is true?

- 1) $(p \rightarrow q) \wedge r$
- 2) $(p \rightarrow q) \wedge \sim r$
- 3) $(p \wedge q) \wedge (p \vee r)$
- 4) $q \rightarrow (p \wedge r)$

15. Which of the following is always true?

- 1) $(p \rightarrow q) \equiv (\sim q \rightarrow \sim p)$
- 2) $\sim (p \vee q) \equiv (\sim p \vee \sim q)$
- 3) $\sim (p \rightarrow q) \equiv (p \vee \sim q)$
- 4) $\sim (p \wedge q) \equiv (\sim p \wedge \sim q)$

16. The negation of the statement “If I become a teacher, then I will open a school” is

(AIEEE-2012)

- 1) I will become a teacher and I will not open a school
- 2) Either I will not become a teacher or I will not open a school
- 3) Neither I will become a teacher nor I will open a school
- 4) I will not become a teacher or I will open a school.

17. The contrapositive and converse of the statements “if the two lines are parallel, then they do not intersect in the same plane” is

- 1) If the two straight lines intersect in a plane, then the lines are not parallel. If the two lines do not intersect in the same plane, then the two lines are parallel
- 2) If the two straight lines intersect in a plane, then the lines are parallel. If two lines do not intersect in the same plane, then the two lines are parallel.
- 3) If two straight lines intersect in a plane, then the lines are not parallel. If the two lines do not intersect in the same plane, then the two lines are not parallel.
- 4) If the two lines do not intersect in the same plane, then the two lines are not parallel. If two straight lines intersect in a plane, then the lines are not parallel.

18. $(p \Rightarrow q) \wedge (q \Rightarrow r) \Rightarrow (p \Rightarrow r)$ is

- 1) a tautology
- 2) a contradiction
- 3) a tautology and a contradiction
- 4) neither a tautology nor a contradiction

LEVEL - III - KEY

- 1) 1 2) 3 3) 2 4) 3 5) 2 6) 4
- 7) 2 8) 2 9) 2 10) 2 11) 1 12) 2
- 13) 4 14) 4 15) 1 16) 1 17) 1 18) 1

LEVEL-III HINTS

1) Let $P : 5 > 7$; $q : 6 < 4$; $\sim (p \rightarrow q) = p \wedge \sim q$

p	q	$\sim q$	$p \leftrightarrow \sim q$	$\sim(p \leftrightarrow \sim q)$	$p \leftrightarrow q$
T	T	F	F	T	T
T	F	T	T	F	F
F	T	F	T	F	F
F	F	T	F	T	T

2)

4). Let p: it is raining q: it is cool

$$\sim (p \wedge q) = (\sim p) \vee (\sim q)$$

5). $p \wedge (\sim q) = \sim (p \Rightarrow q)$; $(\sim p \vee q) = p \Rightarrow q$

hence $[p \wedge (\sim q)] \wedge [(\sim p) \vee q]$ is a contradiction.

$\sim p \Rightarrow q$	$p \Rightarrow q$	$(\sim p \Rightarrow q) \wedge (p \Rightarrow q)$
T	T	F
F	T	F

- 6). $p \wedge (\sim q \vee r) \vee (\sim r \vee q)$
 $p \wedge (q \Rightarrow r) \vee (r \Rightarrow q)$
- 7). p: at least one rational number $x \in S$ such that $x > 0$
 $\sim p$: There is no rational number $x \in S$ such that $x > 0$
 ie. All rational numbers $x \in S$ satisfy $x \leq 0$.
- 9). $x^2 - 3x - \sqrt{2}x + 3\sqrt{2} = 0$
 $x(x-3) - \sqrt{2}(x-3) = 0$
 $\Rightarrow (x-3)(x-\sqrt{2}) = 0 \Rightarrow x = 3$ or $x = \sqrt{2}$,
 $x \in \mathbb{Q} \Rightarrow x \neq \sqrt{2}$
- 11). p: $\triangle ABC$ is isosceles q: the base angles A and B are equal. $\sim(p \Rightarrow q) = p \wedge \sim q$
- 12) $p \rightarrow (q \rightarrow p) = (\sim p) \vee (\sim q \vee p) = (\sim q \vee p) \vee (\sim p)$
 $= (\sim q) \vee (p \vee \sim p) = \sim q \vee t = t$
 $p \rightarrow (q \vee p) = \sim p \vee (p \vee q) = (\sim p \vee p) \vee q$
 $= t \vee q = t$
 $\therefore p \rightarrow (q \rightarrow p)$ is equivalent to $p \rightarrow (p \vee q)$
- 13) p: $x+2=6$; q: $x=4$
 Converse of $p \rightarrow q$ is $q \rightarrow p$; If $x=4$ then $x+2=6$
- 14)
- | p | q | r | $q \rightarrow p$ | $p \wedge q$ | $p \vee r$ | $p \wedge r$ | $q \rightarrow (p \wedge r)$ |
|---|---|---|-------------------|--------------|------------|--------------|------------------------------|
| T | F | T | F | F | T | T | T |
- 16). p: I become a teacher; q: I will open a school
 $\sim(p \Rightarrow q) = p \wedge \sim q$
 I will become a teacher and I will not open a school.
- 17). p: Two lines are parallel
 q: The two lines do not intersect in the same plane.
 Contra positive of $p \Rightarrow q$ is $\sim q \Rightarrow \sim p$
 Converse of $p \rightarrow q$ is $q \rightarrow p$
- 18)

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow r$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	F	F	T	F
T	F	T	F	T	T
F	F	F	T	T	T
F	F	T	T	T	T
F	T	T	T	T	T
F	T	F	T	F	T



LEVEL-IV

1. Observe the following statements:

Statement- I : $(p \vee q) \wedge (\sim p \wedge \sim q)$ is a contradiction.

Statement - II : A statement pattern is called a contradiction, if it is always false, whatever may be the truth values of its constituent statements.

- 1) statement-I is true, statement-II is true, statement-II is a correct explanation of statement-I
- 2) statement-I is true, statement-II is true, statement-II is not a correct explanation of statement-I
- 3) statement-I is true, statement-II is false
- 4) statement-I is false, statement-II is true

2. Observe the following statements

statement-I : $p \wedge \sim(p \wedge q)$ is a tautology

statement-II : A statement pattern is called a tautology, if it is always true, whatever may be the true values of constitute statements.

- 1) statement-I is true, statement-II is true, statement-II is a correct explanation of statement-I
- 2) statement-I is true, statement-II is true, statement-II is not a correct explanation of statement-I
- 3) statement-I is true, statement-II is false
- 4) statement-I is false, statement-II is true

3. Let p be the statement "x is an irrational number", q be the statement "y is a transcendental number" and r be the statement "x is a rational number iff y is a transcendental number". (AIEEE-2008)

Statement-1: r is equivalent to either q or p.

Statement-2: r is equivalent to $\sim(p \leftrightarrow \sim q)$

- 1) S-1 is true, S-2 is a correct explanation of S-1
- 2) S-1 is true, S-2 is true, S-2 is not a correct explanation of S-1
- 3) S-1 is true, S-2 is false
- 4) S-1 is false, S-2 is false

4. **Statement-1** : $\sim(p \leftrightarrow \sim q)$ is equivalent to $p \leftrightarrow q$

Statement-2: $\sim(p \leftrightarrow \sim q)$ is a tautology

(AIE-2009)

- 1) S-1 is true, S-2 is true, S-2 is not a correct explanation for S-1
- 2) S-1 is true, S-2 is false
- 3) S-1 is false, S-2 is false
- 4) S-1 is true, S-2 is true, S-2 is a correct explanation for S-1

5. Consider the following statements

p: Suman is brilliant

q: Suman is rich

r: Suman is honest

The negation of the statement “Suman is brilliant and dishonest if and only if suman is rich” can be expressed as (AIEEE-2011)

- 1) $\sim q \leftrightarrow \sim p \wedge r$
- 2) $\sim(p \wedge \sim r) \leftrightarrow q$
- 3) $\sim p \wedge (q \leftrightarrow \sim r)$
- 4) $\sim(q \leftrightarrow (p \wedge \sim r))$

6. **Statement-I**: $(p \wedge \sim q) \wedge (\sim p \wedge q)$ is a fallacy

Statement-II: $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$ is a tautology. (JEE-Mains 2013)

- 1) statement-I is true, statement-II is true, statement-II is a correct explanation of statement-I
- 2) statement-I is true, statement-II is true, statement-II is not a correct explanation of statement-I
- 3) statement-I is true, statement-II is false
- 4) statement-I is false, statement-II is true

LEVEL-IV - KEY

- 1) 1 2) 4 3) 4 4) 2
- 5) 4 6) 2

LEVEL-IV HINTS

- 1). $(\sim p \wedge \sim q) = \sim(p \vee q)$
 $(p \wedge q) \wedge \sim(p \vee q) = c$

p	q	$p \wedge q$	$\sim(p \wedge q)$	$p \wedge \sim(p \wedge q)$
T	T	T	F	F
T	F	F	T	T
F	T	F	T	F
F	F	F	T	F

2)

p	q	$\sim p \leftrightarrow q$	S-I	S-II
T	T	F	F	T
T	F	T	T	F
F	T	T	T	F
F	F	F	F	T

3)

p	q	$\sim q$	$p \leftrightarrow \sim q$	$\sim(p \leftrightarrow \sim q)$	$p \leftrightarrow q$
T	T	F	F	T	T
T	F	T	T	F	F
F	T	F	T	F	F
F	F	T	F	T	T

4)

- 5). **p**: Suman is brilliant
q: Suman is rich
r: Suman is dishonest
 $p \wedge \sim r \leftrightarrow q$ stands for suman is brilliant and dishonest if and only if Suman is rich its negation is
 $\sim(p \wedge \sim r \leftrightarrow q) = \sim(q \leftrightarrow p \wedge \sim r)$

p	q	$p \wedge \sim q$	$\sim p \wedge q$	$(p \wedge \sim q) \wedge \sim p \wedge q$	$p \rightarrow q \leftrightarrow \sim q \leftrightarrow \sim p$
T	T	F	F	F	T
T	F	T	F	F	T
F	T	F	T	F	T
F	F	F	F	F	T

6)

Statements 1 and 2 are independent

